

1. Approach to Validate Test/Train Data and Split

To build a robust and generalizable model, the validation strategy was strictly designed to respect the chronological nature of the freight market and completely prevent any future data leakage.

Data Splitting Strategy: Standard random splitting methods are inappropriate for this dataset because freight rates are highly dependent on temporal trends. Therefore, the training dataset was ordered chronologically, and a **Time-Series Split (TimeSeriesSplit)** with 5 folds was implemented. This ensures the model is trained exclusively on historical loads and evaluated only on future unseen loads.

Hyperparameter Optimization: To guarantee the integrity of the validation process, the hyperparameter tuning phase (using RandomizedSearchCV for the CatBoostRegressor) was restricted entirely to the first 8 months of the dataset (month ≤ 8). This prevented the optimization process from indirectly learning future market behaviors.

2. Model Comparison and Selection

During the exploratory phase, I evaluated five different algorithms: Ridge Regression, Random Forest, XGBoost, LightGBM, and CatBoost. Using the Time-Series Split mentioned above, I compared their performance based on Negative Mean Absolute Error (MAE).

Result: While simple linear models (such as Ridge) were insufficient for the complex dynamics of freight pricing, **CatBoostRegressor** was chosen over other tree-based alternatives (like LightGBM, which yielded an MAE of around 150) because it achieved a lower and stable error variance across the time folds. Furthermore, its robust handling of engineered categorical features and resilience to overfitting made it the clear choice for the final pipeline.

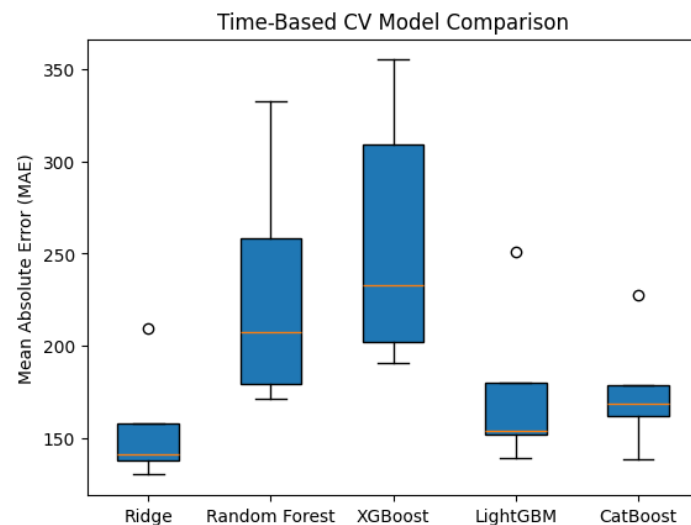


Figure-1: Result of Model Comparison

3. Feature Engineering & Importance Insights

To capture the underlying business logic, several new features were engineered:

- **Geographic Shifts:** `delta_lat` and `delta_lon` to capture the exact vector of movement.
- **Economic Interactions:** `ton_miles` ($\text{weight} \times \text{distance}$) and `market_pressure` ($\text{market_index} \times \text{quote_signal}$) to represent the combined cost and market supply/demand.
- **Temporal Patterns:** An `is_weekend` flag to capture operational pricing differences.

Upon training the final CatBoost model, the **Feature Importance** chart confirmed the success of this approach. While raw distance overwhelmingly dominated the pricing decisions, our engineered features, specifically `ton_miles` and `market_pressure`, emerged as highly critical drivers, proving that the model successfully learned the interactions between weight, distance, and market conditions. Furthermore, its robust handling of engineered categorical features, resilience to overfitting, and inherent ability to suppress noisy or irrelevant features without manual feature elimination made it the clear choice for the final pipeline.

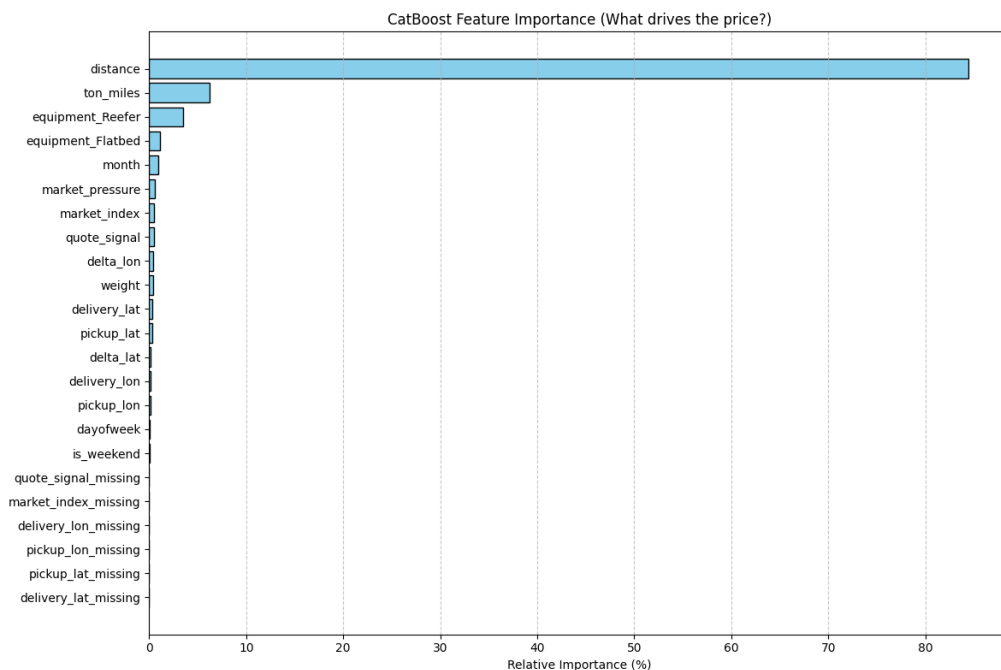


Figure-2: Feature Importance Graph

4. December Prediction Chart

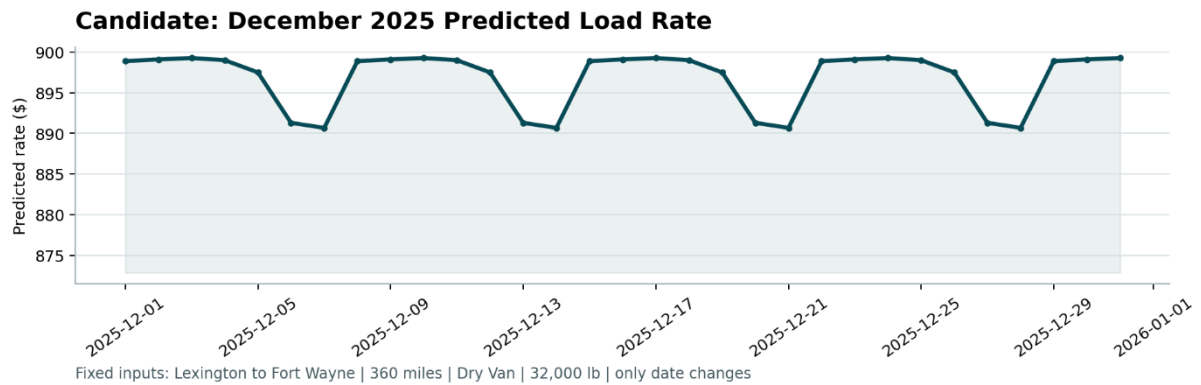


Figure-3: Result of December Prediction Chart Produced by score.py

The December prediction chart demonstrates clear and consistent weekly seasonality corresponding to operational cycles (such as rate dips during weekends when demand drops). Despite keeping all physical parameters constant (Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lbs) and varying only the date, the CatBoost model maintained a stable and realistic price band (\$875 - \$900), proving its strong generalization capability and logical understanding of temporal market trends without erratic fluctuations.